

SHRIRAM CHANDRAN

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EDUCATION

ETH Zürich

Master of Science ETH in Computer Science

since Sep. 2022
Zürich, Switzerland

- **Focus:** Secure & Reliable Systems and Theoretical Computer Science.
- **Thesis:** HO-GDB: Grade 6/6

Indian Institute of Technology Madras

B. Tech. Computer Science and Engineering

Jul. 2018 – Jun. 2022
Chennai, India

- **GPA:** 9.04/10
- **Thesis:** Program Analysis - StarPlat DSL: Grade 9/10

PATENTS AND PUBLICATIONS

- [1] M. Besta^{*†}, S. Chandran^{*†}, J. Cudak, P. Iff, M. Copik, R. Gerstenberger, T. Szyldo, J. Müller, and T. Hoefler, **Higher-order graph databases**, 2025. arXiv: 2506.19661 [cs.DB]. [Online]. Available: <https://arxiv.org/abs/2506.19661>.
- [2] M. Kabić^{*†}, S. Chandran, and G. Alonso, **Maximus: A modular accelerated query engine for data analytics on heterogeneous systems**, New York, NY, USA, Jun. 2025. doi: 10.1145/3725324. [Online]. Available: <https://doi.org/10.1145/3725324>.
- [3] R. Nasre, A. Kumar, E. Rajadurai, N. Behera, R. P. Muniasamy, S. Mohammed, S. Nitish, J. Khatri, N. S. S. Chandran, M. Venkatakrishna, and R. Lalwani, **System and method for automatic parallel code generation for graph algorithms for multiple target architectures**, Indian Patent 432922, Granted, Nov. 2022.

EXPERIENCE

Master Thesis with SPCL

HO-GDB: A Unified Framework for Integrating Higher-Order Structures into Graph Databases

Oct. 2024 - Apr. 2025
ETH Zürich

- Designed algorithms to project higher-order graphs as plain graphs, performance models and proofs of correctness.
- Reasoned about analytical and transactional integrity for higher-order OLTP and OLAP queries.
- Implemented our analysis in HO-GDB, a higher-order graph database with CRUD capabilities.
- Benchmarked HO-GDB across higher-order graph workloads on different datasets and demonstrated scalability.

Practical Work with Systems Group

GPU-Acceleration of the Maximus Database Engine

Oct. 2023 – July. 2024
ETH Zürich

- Implemented GPU support to Maximus, a distributed, scalable and heterogeneous database engine.
- Designed Arrow-compatible data structures to store databases on GPUs for GPU processing on various architectures.
- Integrated GPU-accelerated operators with the existing Maximus interfaces.
- Benchmarked Maximus-GPU on the complete TPC-H query set and demonstrated massive performance gains.

Bachelor Project with Prof. Rupesh Nasre

Program Analysis - StarPlat DSL

Jan. 2022 – Jun. 2022
IIT Madras

- Implemented compiler level optimizations for StarPlat, a graph processing DSL.
- Analyzed and implemented multiple optimizations for the DSL compiler using AST-level analyses.
- Conducted successful testing of the implemented optimizations on multiple large graphs across different algorithms.

- Produced positive performance gains for generated code.

Undergraduate Research with Prof. Raghavendra Rao

Aug. 2021 – Nov. 2021

Stability Analysis

IIT Madras

- Conducted a comprehensive literature review of previous research on stability of instances for various popular algorithms, in particular Bilu-Linial stability amongst other definitions.
- Analyzed stable instances of Christofides' algorithm under various stability conditions.
- Identified improved approximation bounds for stable instances of metric TSP.
- Developed new approximation algorithms for the vertex cover problem for almost-stable instances in random graphs.

Software Engineering Internship

May 2021 – Jul. 2021

Rubrik

Bangalore, India

- Developed a statistics system with the Blobstore team for analyzing backup metadata job performance.
- Designed a functional model for storing job statistics within their respective databases.
- Enabled access to histograms and statistical data to aid in debugging and data analytics.
- Validated the model on live clusters, demonstrating seamless integration with the main codebase.

Software Engineering Internship

Nov. 2020 – Jan. 2021

Godot Media

Bangalore, India

- Designed and constructed an automatic plagiarism checker to search text against documents on the internet.
- Implemented heuristics and parallelization to increase accuracy and reduce algorithm latency.
- Developed a graphical UI to accept data from a user and display results with highlighted matching text.
- Achieved highly accurate plagiarism detection on articles of up to 2000 words in just a few seconds.

SKILLS

Programming: (**Proficient**) C/C++, Python, Scala, Haskell, MPI/CUDA, (**Competent**) OCaml, Java, C#, RISC-V, x86, SQL, Cypher
Frameworks/Tools: Javascript, Django, HTML / CSS, $\text{\LaTeX}$, Git, Coq

SCHOLASTIC ACHIEVEMENTS

JEE: All India Rank **74** in JEE (Advanced) 2018, All India Rank **221** in JEE (Main) 2018

Olympiads: Placed in the top 36 students in India in the 2018 Indian National **Physics**, **Chemistry**, and **Astronomy** Olympiads

KVPY: Two-time Indian Government KVPY fellow (All India Rank **72** in 2016, **68** in 2017)

SELECTED COURSE PROJECTS

GPU-Acceleration of an Algorithm

Design of Parallel & High-Performance Computing, Sep. 2023 – Dec. 2023

- Implemented a banded matrix inversion algorithm, and performed parallelization using MPI and CUDA to improve the performance of the algorithm.

Building a Program Verifier

Program Verification, Apr. 2023 – Jun. 2023

- Designed and implemented a program verification system to translate programs from EVT, an event-based research language, to Viper, a verification backend language, allowing users to attach specifications with EVT programs.

System Level Optimization of an Algorithm

Advanced Systems Lab, Mar. 2023 – Jun. 2023

- Implemented the PolyVest approximation algorithm for volume estimation of a convex polytope and performed system optimizations including vectorization and cache performance improvement to get more than a 23x speedup.

Nand2Tetris

Computer System Design, Aug. 2019 – Nov. 2019

- Designed a Hack micro-architecture using the Hack Hardware Simulator on Nand2Tetris and implemented an assembler and a compiler for the Jack language.